

Hazard Identification and Risk Assessment (HIRA)

Duration: 1 Day

Mode: Online / Classroom / On-Site

Certification: QLSS Certificate of Completion



Course Introduction

Hazard Identification and Risk Assessment (HIRA) is a proactive safety management tool used to identify potential workplace hazards, assess the associated risks, and determine appropriate control measures. HIRA is essential for compliance with Occupational Health and Safety (OHS) regulations, ISO 45001:2018, and various industry-specific safety standards.

This training equips participants with practical tools to conduct HIRA, calculate risk levels, apply the risk matrix method, and implement control strategies aligned with the hierarchy of controls. Real-world examples from manufacturing, construction, and maintenance operations are used to build practical competency.



Pre-requisites

- Basic understanding of workplace operations or safety practices
- No prior safety certification required
- Ideal for team leads, safety officers, and engineers



Learning Objectives

- Understand the fundamentals of hazard identification and risk assessment
- Apply a structured approach to identifying physical, chemical, mechanical, and ergonomic hazards
- Calculate risk using probability-impact matrices
- Apply the hierarchy of control to mitigate identified hazards
- Prepare and maintain HIRA registers
- Support safety compliance with ISO 45001 and industry norms



Course Agenda

1. Introduction to HIRA

- Definition and purpose of HIRA
- Legal and standard-based requirements (e.g., ISO 45001, Factories Act)
- Role of HIRA in incident prevention and operational safety

2. Hazard Identification Techniques

- Types of hazards: physical, chemical, biological, ergonomic, psychosocial
- Sources of hazard: machine, material, method, man
- Workplace inspection, task-based analysis, historical data

3. Risk Assessment Principles

- Risk = Likelihood × Severity
- Risk matrix: 3x3 and 5x5 formats
- Classifying risks: Low / Medium / High / Unacceptable

4. Control Measures and Action Planning

- Hierarchy of controls: Elimination to PPE
- Engineering, administrative, and behavioral interventions
- Assigning risk owners and timelines for mitigation

5. Preparing a HIRA Register

- Format, elements, and documentation
- Integrating with Safety Management Systems
- Updating HIRA post-incident or change

6. Compliance and Audit Preparedness

- ISO 45001:2018 clause mapping
- Aligning with internal and external audit requirements
- Sample findings and improvement actions



What You'll Receive

- QLSS Certificate of Completion
- Training Material / Handout (PDF Format)
- Final Assessment with Individual Evaluation Report



Who Should Attend

- Safety Officers and EHS Managers
- Engineers and Line Supervisors
- Maintenance and Facility Teams
- Project Coordinators and HSE Leads
- Internal Auditors and Risk Committee Members



Trainer Profile

Our trainers are certified HSE professionals with experience in risk assessments across heavy industry, infrastructure, automotive, and pharma sectors. Their sessions focus on hands-on hazard identification, risk scoring, and control implementation with relevance to shopfloor, office, and project environments.



Instructional Methodology

- Visual walkthrough of HIRA registers
- Real-world hazard mapping from images and videos
- Small-group exercises with risk scoring
- Case-based discussion and lessons-learned sharing
- Compliance focus with ISO 45001:2018 and statutory rules



What Makes QLSS HIRA Training Unique

- Field-oriented hazard examples from Indian industries
- Focus on both documentation and behavioral risk factors
- Structured templates and risk matrix tools
- Applicable to ISO 45001, OHSAS, and factory safety rules
- Expert-led guidance on audit preparedness and compliance